# Vishal Kumar

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## Education

**National Institute of Technology, Jamshedpur** — Jharkhand, India
*B.Tech. in Electrical Engineering | CGPA: 8.55* — *Aug 2024 – Jul 2028*

**Govt. Saryug Singh Baldeo Singh Higher Secondary School, Mehura** — W. Champaran, Bihar
*AISSCE – PCM | Percentage: 87.4%* — *Jun 2022 – Mar 2024*

**J N V VRINDAVAN WEST CHAMPARAN BIHAR** — Bihar, India
*Matriculation | Grade : 95.2%* — *May 2022*

## Projects

**Plags Checker** | *JavaScript, React.js, Tailwind CSS, Node.js, Vitest* | Live Demo | Source Code — *Jul 2026*

- Built a C++ similarity engine using lexical tokenization, normalization, n-grams, and JavaScript.
- Applied winnowing fingerprints and weighted scoring to detect renaming, dead code, and statement reordering.
- Created Vitest benchmarks to validate detection across source-code transformations and distinct algorithms.

**AI Chatbot** | *React.js, Node.js, Express.js, MongoDB, Gemini API, Tailwind* | Live Demo | Source Code — *May 2026*

- Developed a MERN AI chat platform with REST APIs, and user-specific chat management.
- Integrated Gemini API with Markdown and code-block rendering to deliver structured AI responses.
- Secured backend routes using JWT, HTTP cookies, middleware, protected routes, and user authorization.

**Developer Portfolio** | *React.js, Express.js, REST APIs, Framer Motion* | Live Demo | Source Code — *May 2026*

- Designed a full-stack portfolio using React, Express.js, MongoDB, and REST APIs.
- Implemented cached APIs for live LeetCode and Codeforces statistics with automated data synchronization.
- Crafted reusable React components with responsive design and Framer Motion animations.

## Skills

- **Languages**: C, C++, JavaScript
- **Web**: HTML, CSS, React.js, Tailwind CSS, Node.js, Express.js, MongoDB, REST APIs, JWT Authentication
- **Tools**: Git, GitHub, Postman, Vercel
- **CS Fundamentals**: Data Structures and AlgorithmsDSA), Operating Systems(OS), OOPs, CN

## Achievements

- Secured **3rd place** in a competitive programming contest organized by the Electrical Engineering Society, NIT Jamshedpur.
- Ranked among the top **1.2%** of 1.5 million candidates in JEE Main 2024 and qualified for JEE Advanced.
- **LeetCode**: Max. Rating – **1746** | Profile
- **Codeforces**: Max. Rating – **1279**| Profile
- **CodeChef**: Max. Rating – **1390** | Profile

## Leadership

**Volunteer Educator, SANKALP (NGO)** — *Nov 2024 – Apr 2026*

- Mentored underprivileged students through free structured academic support and regular learning sessions.
- Collaborated on community education initiatives through academic guidance and mentoring.

**Captain, U-14 Kho Kho Team** — *Jul 2019 – Sep 2020*

- Led the U-14 Kho Kho team and coordinated practice sessions, match preparation, and on-field strategy.
- Guided the team to regional qualification through leadership, teamwork, and consistent training.